

Notes: Coimbra & Rey (RES, 2024)

Financial Cycles with Heterogeneous Intermediaries

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1 Big picture

- **Question.** How can heterogeneity among financial intermediaries generate financial cycles, systemic risk, and a risk-taking channel of monetary policy?
- **Core departure from representative-intermediary models.** The financial sector contains a cross-section of intermediaries with different Value-at-Risk (VaR) constraints and different risk-taking behaviour.
- **Main mechanism.** When funding costs fall, risky intermediaries expand their balance sheets more aggressively. Assets move toward the balance sheets of the most risk-taking intermediaries.
- **Financial-cycle force.** Low interest rates relax funding costs and encourage leverage, entry, and risk shifting. This can stimulate investment but also concentrate risk in fragile balance sheets.
- **Non-linearity.** Monetary stimulus has different effects depending on the initial level of interest rates. At high rates, lower rates can stimulate activity and reduce aggregate risk. At low rates, further easing can increase financial instability.
- **Risk premium result.** In the risky steady state, monetary easing can raise systemic risk while lowering the risk premium. This creates a trade-off between macroeconomic stimulus and financial stability.
- **Stylized facts.** Balance-sheet data show that leverage dynamics are heterogeneous across intermediaries, with the most levered and largest intermediaries responding most strongly to lower funding costs.
- **Policy insight.** A monetary policy expansion can produce a “good boom” or a “bad boom.” The difference depends on whether easing primarily brings safer intermediaries into markets or strengthens already risky intermediaries.
- **Takeaway.** Financial stability is not only about aggregate leverage. It also depends on the cross-sectional allocation of assets toward intermediaries with loose risk constraints.

2 Data and measurement

2.1 Balance-sheet data and leverage

- The empirical stylized facts use Bankscope balance-sheet data for financial intermediaries.
- Leverage is defined as assets over equity at book value:

$$Lev_{i,t} = \frac{Assets_{i,t}}{Equity_{i,t}}. \quad (1)$$

- The paper studies asset-weighted leverage quantiles, unweighted leverage, and equity quantiles.
- The sample focuses on cross-sectional heterogeneity: top quantiles, medians, and lower quantiles behave differently over the financial cycle.

Interpretation: The empirical section is designed to show that leverage is not a representative-intermediary object. Risk builds up unevenly in the cross-section.

2.2 Interest rate exposure

- Funding costs are proxied by the real or nominal effective Federal Funds Rate.
- The paper studies how leverage and the asset share of highly levered intermediaries vary with interest rates.
- The key empirical pattern is that top-leverage intermediaries are more sensitive to lower funding costs.

Interpretation: The fact that the most levered intermediaries respond most to interest rates motivates a model in which monetary policy affects both aggregate investment and the cross-sectional allocation of risk.

2.3 Panel regression for heterogeneous leverage sensitivity

The baseline panel regression is

$$Lev_{i,t} = \theta_0 + \theta_1 Lev_{i,t-1} + \theta_2 FF_t + \theta_3 Top5_{i,t} + \theta_4 (FF_t \times Top5_{i,t}) + \alpha_i + \varepsilon_{i,t}. \quad (2)$$

- $Top5_{i,t}$ equals one if intermediary i is in the top 5% of the leverage distribution.
- FF_t is the real Federal Funds Rate.
- θ_4 measures whether highly levered intermediaries respond differently to interest rates.

Interpretation: A negative interaction means that top-leverage intermediaries increase leverage more when funding costs fall.

2.4 Systemic risk measures

The model tracks multiple measures of financial instability:

- the cut-off VaR type α_t^L separating safe and risky intermediaries;
- leverage skewness and cross-sectional concentration;
- the weighted option value of default;
- funding spreads and premia over deposits.

Interpretation: Systemic risk is a distributional object. The location of assets and leverage across intermediary types matters as much as aggregate leverage.

3 Model primitives

3.1 Technology and intermediaries

- The economy has households, financial intermediaries, and a risky production technology.
- Intermediaries are heterogeneous in their VaR parameter α_i .
- A looser VaR constraint means that the intermediary can tolerate a higher probability of failing to repay stakeholders.
- Intermediaries decide whether to participate in risky capital markets, whether to borrow deposits, and how much to lever.

Interpretation: Heterogeneity in α_i is the model's main state of the financial sector. It determines which intermediaries enter, which lever, and where assets are held.

3.2 Intermediary balance sheet and objective

Intermediary i chooses investment in risky assets $k_{i,t}$, storage $s_{i,t}$, and deposits $d_{i,t}$. The balance sheet is

$$k_{i,t} + s_{i,t} = \omega_{i,t} + d_{i,t}, \quad (3)$$

where $\omega_{i,t}$ is intermediary equity.

The intermediary maximizes expected next-period consumption:

$$V_{i,t} = \max E_t(c_{i,t+1}). \quad (4)$$

Net cash flow is

$$\pi_{i,t+1} = R_{i,t+1}^K k_{i,t} + s_{i,t} - R_t^D d_{i,t}, \quad (5)$$

and limited liability implies

$$c_{i,t+1} = \max\{0, \pi_{i,t+1}\}. \quad (6)$$

Interpretation: Limited liability creates a risk-shifting motive. Intermediaries enjoy upside returns but do not fully internalize downside losses.

3.3 VaR constraint

The intermediary faces

$$\Pr(\pi_{i,t+1} < \omega_{i,t}) \leq \alpha_i. \quad (7)$$

- α_i is the maximum allowed probability of not repaying stakeholders fully.
- Lower α_i means a tighter constraint.
- Higher α_i means the intermediary can take more risk.

Interpretation: The VaR constraint is the formal source of heterogeneous risk-taking. Different intermediaries behave differently even when they face the same aggregate prices.

3.4 Limited liability and default option

Limited liability implies

$$E_t[\max(0, \pi_{i,t+1})] \geq E_t[\pi_{i,t+1}], \quad (8)$$

with a strict inequality when default has positive probability.

Deposit insurance transfers are

$$tt_{t+1}^i = \max(0, -\pi_{t+1}^i). \quad (9)$$

Interpretation: The value of default is an option value. It is larger for intermediaries with looser VaR constraints and larger balance sheets.

4 Equilibrium and theoretical mechanisms

4.1 Extensive and intensive margins

- **Extensive margin:** whether an intermediary participates in risky capital markets and whether it borrows deposits.
- **Intensive margin:** how much a risky intermediary levers up, conditional on borrowing.
- If the expected return on risky capital is high enough, a participating intermediary either does not borrow or borrows up to its VaR constraint.

Interpretation: This all-or-nothing result makes the cross-section sharp. Risky intermediaries operate at their constraints.

4.2 Leverage formula

For an active intermediary with binding VaR constraint, leverage is

$$\lambda_{i,t} = \frac{k_{i,t}}{\omega_{i,t}} = \frac{r_t^D}{r_t^D - r_t^{\alpha_i}}, \quad (10)$$

where $r_t^{\alpha_i}$ is the return on capital that makes the risky intermediary's return on equity equal to zero.

Interpretation: Lower deposit rates increase leverage, especially when r_t^D is already close to the risky intermediary's break-even return.

4.3 Cut-off intermediary

There exists a cut-off α_t^L such that

- intermediaries with $\alpha_i < \alpha_t^L$ have tighter constraints and do not lever;
- intermediaries with $\alpha_i \geq \alpha_t^L$ lever up to their VaR constraints;
- the cut-off is endogenous and moves with deposit rates and expected productivity.

Interpretation: Monetary policy changes both how much existing risky intermediaries lever and which intermediaries enter the risky leveraged sector.

4.4 Good booms and bad booms

When interest rates fall from a high level:

- entry by previously safe intermediaries can dominate;
- aggregate investment rises;
- the average default probability can fall;
- financial stability may improve.

When interest rates fall from a low level:

- already-risky intermediaries expand balance sheets more aggressively;
- risk concentrates in the most levered institutions;
- leverage skewness and weighted default option values rise;
- the risk premium can fall even as systemic risk rises.

Interpretation: The model delivers a non-linear risk-taking channel of monetary policy: the same policy move can be stabilizing or destabilizing depending on the initial state.

4.5 Costly default and crises

In the extension with costly default, assets held by defaulting intermediaries suffer productivity losses. The aggregate loss is increasing in the share of capital held by defaulting intermediaries.

Interpretation: When risky intermediaries control a large share of assets, a productivity shock can generate a deep crisis with high funding costs, low output, and persistent wealth effects.

5 Identification and empirical interpretation

5.1 What the stylized facts support

- Leverage is convex in asset size but not in equity size.
- The top 5% of levered intermediaries account for a large and cyclically changing share of assets.
- The asset share of the top-leverage group is negatively related to the Fed Funds Rate.
- Highly levered intermediaries are more interest-rate-sensitive than other intermediaries.

Interpretation: These facts justify a heterogeneous-intermediary model. A representative intermediary would miss the cross-sectional concentration of risk.

5.2 What the calibration shows

- The model is calibrated to match leverage and default probabilities for risky intermediaries.
- The distribution of VaR parameters determines which intermediaries lever and how risk is allocated.
- Monetary shocks and productivity shocks generate different financial-cycle patterns.

Interpretation: The calibration is qualitative and mechanism-oriented. It is meant to show how heterogeneity and entry produce financial-cycle dynamics.

5.3 Limitations

- The model abstracts from many institutional details of banks and shadow banks.
- VaR heterogeneity is exogenous.
- Deposit insurance and default costs are stylized.
- The empirical facts are stylized rather than causal estimates.

Interpretation: The paper's strength is the mechanism linking heterogeneity, monetary policy, and systemic risk, not a single micro-identified causal estimate.

6 Tables and figures

The figures and table below are directly cropped from the paper and grouped compactly.

6.1 Figures 1 and 2: heterogeneity in intermediary leverage

Read:

- Figure 1 shows that asset-weighted leverage rose sharply in top quantiles before the crisis but not in lower quantiles.
- The top leverage quantiles behave very differently from the median and bottom quantiles.
- Figure 2 shows leverage is convex in asset size: larger intermediaries are more levered.
- The relationship with equity deciles is much weaker.

Economic message: Leverage and risk-taking are concentrated among large intermediaries. This motivates the model's focus on cross-sectional heterogeneity.

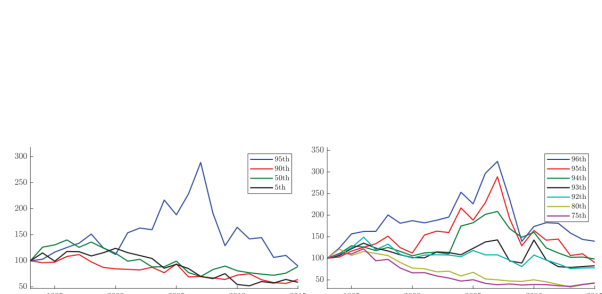


FIGURE 1
Evolution of asset-weighted leverage by quantiles (base year = 1993)
Note: Quantiles of each year's cross-sectional distribution of asset-weighted leverage, rebased to 100 at the beginning of sample. Asset-weights calculated by dividing balance sheet size by the respective yearly mean balance sheet size.

Figure 1: asset-weighted leverage by quantiles

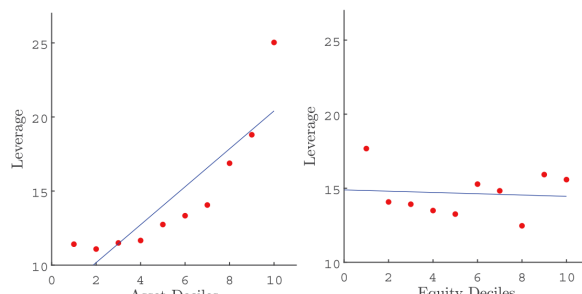


FIGURE 2
Mean leverage by asset deciles (left) and equity deciles (right)
Note: Binned scatter plots with intermediaries grouped in the x-axis into 10 bins by balance sheet size (left panel) and equity (right panel). Dots represent the mean leverage (unweighted) of each group. Each bin contains an average of 41 distinct financial intermediaries and

Figure 2: leverage by asset and equity deciles

6.2 Figures 3 and 4: interest-rate sensitivity of risk concentration

Read:

- Figure 3 shows a negative relation between the asset share of the top 5% most levered intermediaries and the real or nominal Fed Funds Rate.
- Figure 4 shows that higher leverage percentiles are more sensitive to interest-rate changes.
- Equity percentiles do not display the same robust pattern.

Economic message: Lower funding costs shift assets toward already highly levered intermediaries. This is the empirical motivation for the risk-taking channel.

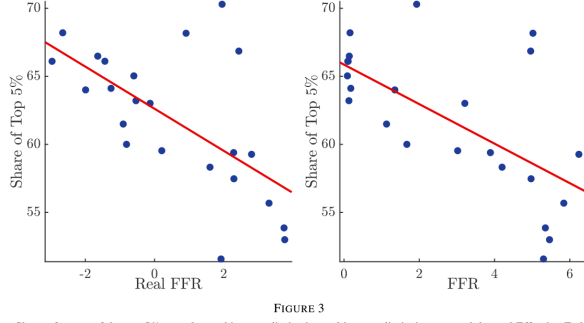


FIGURE 3
Share of assets of the top 5% most levered intermediaries in total intermediaries' assets and the real Effective Fed Funds Rate (left panel) and the nominal effective Fed Funds rate (right panel) in pp

Figure 3: top-5% leverage share and Fed Funds Rate

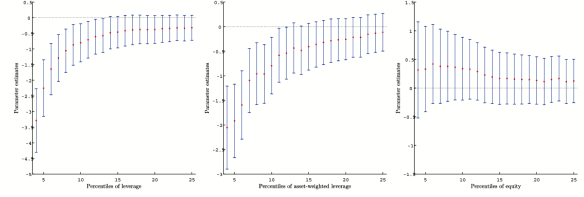


FIGURE 4
Interaction coefficients under different percentile choices

Figure 4: interaction coefficients across percentile choices

6.3 Figures 5 and 6: model cross-section and non-linear margins

Read:

- Figure 5 shows cross-sectional leverage as a function of the VaR parameter and asset quantiles.
- Low interest rates increase balance-sheet expansion among risky intermediaries and raise dispersion.
- Figure 6 shows how the cut-off α_t^L , aggregate capital K , and leverage skewness vary with deposit rates.
- The cut-off and skewness are non-linear in funding costs.

Economic message: Monetary policy affects both the intensive margin of leverage and the extensive margin of entry into risky intermediation.

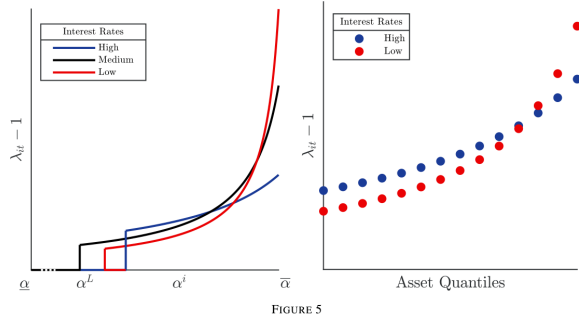


FIGURE 5
Cross-sectional distribution of leverage λ_{it} as a function of the VaR parameter α^L (left panel) and quantiles of asset size among leveraged institutions (right panel). $\lambda_{it} - 1$ is used so that unlevered participants have a value of 0.

Figure 5: leverage distribution in the model

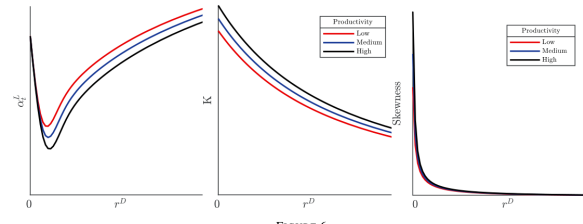


FIGURE 6

Cut-off level α_t^L and aggregate capital stock as a function of deposit rates r_t^D

Figure 6: cut-off, capital, and skewness

6.4 Figure 7 and Table 1: partial-equilibrium IRFs and calibration

Read:

- Figure 7 shows the partial-equilibrium effect of a 100 basis point fall in deposit rates.
- Output rises sharply, while the cut-off and concentration effects depend on the initial interest-rate environment.
- Table 1 reports the key calibrated parameters, including risk aversion, discount factor, productivity parameters, spread persistence, equity of intermediaries, and the VaR distribution.

Economic message: The calibration is designed to match a financial system in which low funding costs can stimulate output while also concentrating risk.

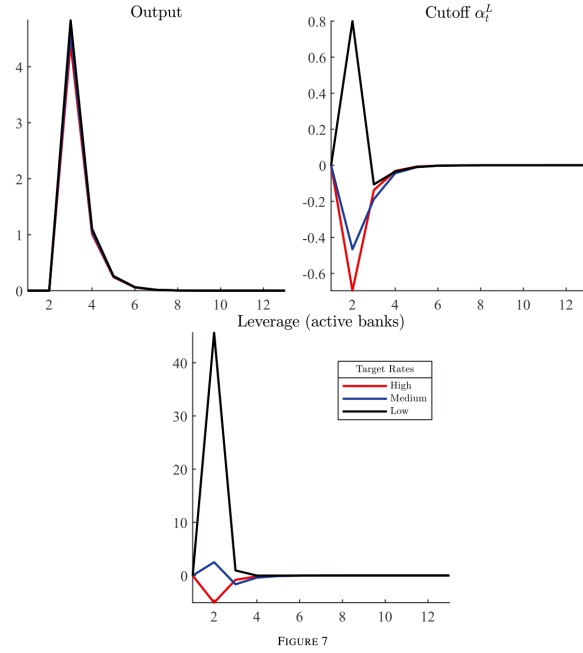


FIGURE 7
Partial equilibrium IRF to a 100 basis points fall in deposit rates. Scale in percentage point deviations from the baseline

Figure 7: partial-equilibrium IRFs

Calibration of selected parameters		
Parameter	Value	Description
ψ	4	Risk aversion parameter
β	0.96	Subjective discount factor
ρ^z	0.9	AR(1) parameter for TFP
σ_z	0.03	Standard deviation of TFP shock
μ_r	0.023	Target spread over deposit rates
ρ_r	0.816	Spread persistence
σ_r	0.0128	Standard deviation of spread
$\frac{\lambda}{1+\lambda}$	0.41	Wholesale funding percentage
θ	0.35	Capital share of output
δ	0.1	Depreciation rate
ω	0.697	Equity of intermediaries
$\bar{a}$	0.4961	Upper bound of distribution $G(a^i)$
ζ	0.01	Idiosyncratic unproductive capital probability

Table 1: calibration of selected parameters

6.5 Figures 8 and 9: monetary policy shock

Read:

- A 100 basis point monetary policy shock raises output and affects the cut-off, active-bank leverage, funding rates, premia, and the weighted option value of default.
- In the risky steady state, the weighted option value of default rises strongly.
- Funding rates and premia can fall even as systemic risk rises.

Economic message: The model delivers a risk-taking channel of monetary policy: easing can reduce risk premia while increasing systemic risk.

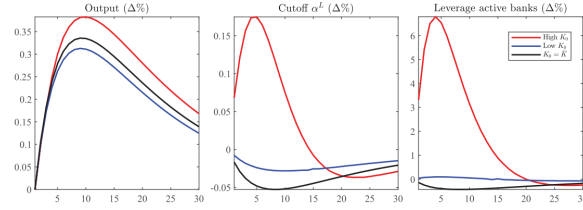


FIGURE 8
Monetary policy shock of 100 basis points to γ_t

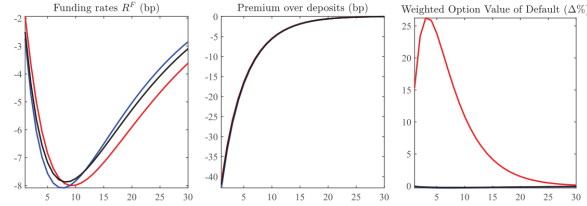


FIGURE 9
Monetary policy shock of 100 basis points to γ_t : financial variables

a 100 basis points increase to the subsidy³⁹ in three different scenarios to illustrate the non-linear effects of monetary policy on financial stability. Impulse response functions are expressed as deviations from the respective scenario in the absence of the shock. This monetary policy

6.6 Figures 10 and 11: productivity shocks and costly default

Read:

- Figure 10 shows that a standard productivity shock produces a “good boom”: leverage and the premium move only mildly.
- Figure 11 shows that a large productivity shock with costly default produces large output losses and sharp funding-rate increases.
- The crisis is worse when defaults propagate or when assets are concentrated in risky intermediaries.

Economic message: Credit-supply-driven booms differ from productivity-driven booms. Concentration of risk in risky intermediaries makes negative shocks more costly.

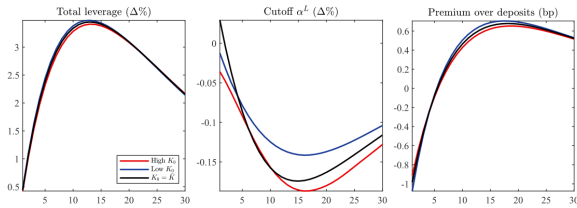


FIGURE 10
Shock to exogenous productivity

Figure 10: standard productivity shock

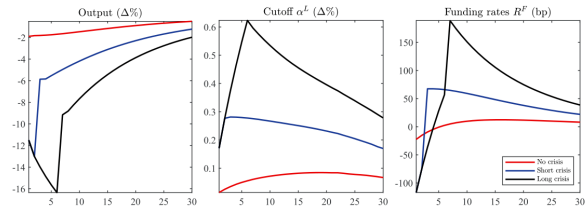


FIGURE 11
Large shock to exogenous productivity

Figure 11: large productivity shock with default

7 Short summary

- The paper builds a dynamic macro-finance model with heterogeneous financial intermediaries.
- Intermediaries differ in VaR constraints, which generate heterogeneous leverage and risk taking.
- The empirical stylized facts show that leverage is concentrated among large, highly levered intermediaries and is sensitive to interest rates.
- In the model, falling deposit rates affect both entry into risky intermediation and leverage of already-risky intermediaries.

- At high interest rates, monetary easing can increase output and improve financial stability.
- At low interest rates, monetary easing can increase output while worsening financial stability.
- The risk premium can fall even as systemic risk rises.
- Costly intermediary default makes the distribution of assets across intermediaries macroeconomically important.
- The main lesson is that monetary policy affects not only aggregate credit but also who holds risk.

8 Conclusion

8.1 Core conclusion

Coimbra and Rey show that financial cycles can arise from the interaction between heterogeneous risk-taking intermediaries and funding conditions. Low interest rates can stimulate investment, but when rates are already low, further easing can shift assets toward the most risk-taking intermediaries and increase systemic risk.

8.2 Economic intuition

The key intuition is that not all intermediaries react equally to funding costs. Intermediaries with looser VaR constraints exploit limited liability and expand more aggressively when borrowing is cheap. As their balance sheets grow, financial stability depends less on average leverage and more on the concentration of assets in fragile balance sheets.

8.3 Takeaway

The paper provides a theory of the risk-taking channel of monetary policy. Policy can lower risk premia and raise output while simultaneously increasing fragility. The distribution of leverage across intermediaries is therefore a central state variable for macroeconomic and financial stability analysis.